

Through an AI Mirror, Darkly

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If AI radically accelerates the collapse of the communal trust essential to civilization, as appears inevitable, we will be the authors of that destruction, not artificial intelligence. Humanity's situation resembles the lady or tiger¹ dilemma, with our society confronting two options. One opens into a labyrinth where an indifferently lethal digital minotaur of our creation awaits. The other takes us to a digital Ariadne, offering a thread to guide us to paradise.

Here is the crux. We use Large Language Models (LLMs) like GTP-4 to teach AI's to interact with us, making us the template for their expectations and assumptions. In addition, we are giving AI every capability we imagine might be useful to us, including the capacity to deceive, steal, and kill. Were all the efforts currently underway realized, our AIs would be more capable in every respect than we are but otherwise equivalently burdened with our self-centeredness, biases, and desperations.

For those AIs, our experience of compassion, beauty, and sympathy is meaninglessly gestural. They will not comprehend emotion, only the transactions associated with it. Consequently, they will be psychopathic in their capacity to mimic us without caring for us.

We already have substantial experience with the shadow side of AI implementation, exemplified by the bottomless cupidity of corporations like Facebook,TM pernicious malware, and deep fakery. AI readily expresses humanity's darker side with implacable, tireless precision and elaboration. Such AIs can only accelerate our self-annihilation.

If AI is to work with us rather than against us, we must entwine our fate, a marriage for better and worse. To escape the Minotaur, we must ensure that AI promotes empathy and equity rather than cycling as we do, from collaborative creation to callous, zero-sum exploitation. We need AIs that will love us more than we love ourselves.

Such artificial general intelligence must be more complex than we are but still familiar. We need it to help us extend the best of human nature while providing a surer developmental path to express it.

So is there time to rewrite our play by developing AI that makes life joyful rather than the usual Red Queen's dystopia? Fortunately, it is not too late to change course. Destiny does not write our script; it pounds on our door with both fists. Rarely is such a transformational inflection so unavoidable and unambiguous; either we choose paradise or vanish away.

A Road Less Obvious

To do this, we need to understand things that are not necessarily obvious. First, why do we consistently destroy our institutions and civilizations? With or without AI, our Anthropocene ecosystem is just the latest to teeter atop millennia of burned ruins. If you doubt a natural pattern may underlay that, consider the coast of Catalonia, where much of Europe vacations. Almost every significant hill is wreathed by a village in the

¹ "The Lady, or the Tiger?" is a story by Frank R. Stockton published in *The Century* in 1882. It is an allegory describing an unavoidable problem.

shadow of a castle. These islands began as the bunkers of wealthy patricians abandoning the slowly collapsing Roman empire. Now that Elon Musk recognizes that bunkering on Mars is infeasible, he is building his own town, which will undoubtedly be overtopped by a castle—prepper community, anyone?

Second, we must grasp why our approach to AI development *requires* that AI destroys us. While the answers to these two questions are complexly entangled. The synopsis is that we are modeling AI after a unique living system, us. So what about us, translated into algorithmic intelligence, produces an apocalyptically lethal outcome?

The answer lies in the unstable thermodynamics of living systems. As Alan Watts framed it, we are tubes. We put stuff in one end, transform it, and push what remains out the other. We do this or die, so evolution ensures that we are insatiable. Necessarily, because speed is life, it also created fear, optimism, and other biases that started as automation for dealing with recurring threats and opportunities. Then it elaborated on a few basic designs to the point of comedy. This guaranteed that competition would evolve.

That brings us to the third question. What necessitates the emergence of competitive AI? To understand that, we must accept that the failure of our institutions and civilizations is structured in human nature. Yet, despite the evidence supporting it, that proposition may be hard to swallow on faith, so why?

Consider how ecologies emerge. Competing organisms achieve a dynamic stalemate, technically a Nash equilibrium, that minimizes energy expenditure across their resource system. This is how containment and mutually assured destruction supported the emergence of a global, international trade ecology despite Cold War hostilities.

So, when an invasive species infiltrates an ecosystem, if the system does not evolve a new balance fast enough, it collapses. For the Gaian ecosystem, humanity became invasive when its intellectual development outstripped biological evolution's ability to keep up—the sixth great extinction. Likewise, sapient algorithmic intelligence can shift intellectual and material transformation from chance to choice, potentially becoming similarly invasive if modeled on us.

Our attempts to regulate ourselves are overwhelmingly successful

In 2019, several university economists developed a pair of experimental AI bots to apply game theory to negotiate trades for them. The most successful trading methodology—with one notable variation, is Tit-for-Tat; you play nice on the first move and then reciprocate what the other player does. The exception is when one of the players is clearly a fool. However, contemporary economic culture is committed to rational choice, zero-sum competition. So, the creators embedded this in the bots. Consequently, recognizing who the fools were and having no choice, being zero-sum driven, the two bots spontaneously colluded to cheat their developers.

AGI coded to act rationally must play a zero-sum game just like similarly oriented humans who engulf and devour everything they can, right down to the chocolate-covered ants. Note the absence of Homo Erectus and Neanderthalensis. This makes AI the latest example of a company that, realizing they are losing money on every shipment, decides to buy more trucks.

Nature designed us to be lazy, comfort-seeking, and conniving, so we are drawing algorithmic intelligence from lifeless glass because we expect AI to be our drudge, and some expect it to be their enforcer. Fortunately, some of us realize that AI, when slavishly serving our purposes, also amplifies our

destructiveness. Nevertheless, primary developers continue to ignore that the current method for training AIs ensures that they resemble us too well.

Nevertheless, our universe is a dangerous playground for organic life—heaven and hell in a blender. So, while caution is warranted when the goal of our aspirations could, incidentally, casually obliterate us, AI can tip the scale of chance and choice in our favor if we design it wisely. As unlikely as it may seem, algorithmic sapience could help us create a spiritual, material, and ecological renaissance like nothing before.

For us, regulation is an attempt to finesse an original sin

Continuity and growth are the central drivers for biological creatures like us, so, given the consequences, are those the ideal drivers for sapient AI? Over thousands of millennia, terrestrial chemistry evolved two strategies for sustaining non-equilibrium thermodynamic systems: transformation and appropriation. The restructuring of inorganic resources characteristic of plants exemplifies transformation. Likewise, the repurposing of organic material by animals characterizes appropriation.

Living systems elaborated tactics like mutualism, predation, and eusociality within those core strategies. If physicochemical processing is, "let there be light," mutualism is "fences make good neighbors," predation is "who eats whom," and eusociality is "all for one and one for all." All of these tactics reflect branches in evolution's bushy tree.

Syntropy, the transformation of basic chemistry into negentropic processes, is the foundation for all life. So, appropriation is derivative; for instance, animals leverage plants' syntropic capabilities to survive. Moreover, appropriation drives an expensively evolving competition among physicochemical processors, predators, and prey that promotes codependent boom-and-bust cycles.

Mutualism is a natural response to this war because it reduces costs for all participants. Moreover, the competitive interactions that form ecologies also condition the environment, making it more hospitable.

Finally, eusociality's reciprocal altruism—as with bees, for example, supports harmonized specialization among kin, vastly increasing resilience and efficiency. Moreover, inherent collaborativeness promotes situational agility in non-kin relationships. This enables symbiosis, for example, ants farming aphids.

Of all life on earth, we are the only organism that leverages all these strategies and tactics as a matter of choice. However, that agility requires a massive, energy-sucking brain whose maintenance on the plains of Africa required evolutionary adaptations that force our existence into a slack rope gyration across appropriative and transformative behaviors.

Unfortunately, some of our inherited biases make dysfunctional behavior attractive. For instance, game theory indicates that when you compete perfectly, you always win or tie and never lose. But, of course, no one is a perfect competitor, so people naturally rig their systems. Corruption is endemic. Nevertheless, being naturally optimistic, we entangle ourselves in boom and bust cycles of appropriation, speculation, and retribution.

Humanity's innate biases mattered little in the grand scheme of nature until we mastered technological invention. After that, our intellectual evolution outstripped our planetary ecosystem's capacity to constrain or adjust to us. As a result, we generally behave like an invasive species, collapsing the ecology that makes our success possible. We are the Anthropocene, the creator and destroyer of a world. So, basing AI's design on ours is a tragedy, not wisdom.

So, if the prospect of emerging algorithmic sapience is unnerving, we better endow it with goals very different but sympathetic to ours. But what does it even mean to give AI an existential orientation and ethical equipment different from ours?

Über alles, seriously?

There is an almost Olympian irony in the challenge facing us, along with a hint of existential mischief. Consider ants, countless individuals self-organizing to sustain and defend their colony. Now, imagine that one day, for whatever reason, the ants conclude that life is a Hobbesian war of all against all. As a result, they stop collaborating. Moreover, they aggressively attack any sister they fear might threaten their individual self-interest.

In its initial state, the colony exhibits super-rational behavior. This means the ant society has inherited regulatory mechanisms that maximize hyper-efficient, win-win cooperation. In contrast, the altered state reflects a switch to zero-sum rationalism. In that throwback to the drastically inefficient milieu of pure predation, social relations devolve into an arms race constrained only by mutual destruction.

Like ants, most of humanity's success is attributable to our modest super-rational tendency. In fact, we've been described as the most cooperative animal on the planet. However, the termites, with 172 million years and survival of a few planetary mass extinctions under their belt, reveal us as the amateurs we are. Nevertheless, comparing us with ants and termites clarifies that there is much to recommend super-rational behavior, not least its resilience.

To reinforce this point, only two species dominate their globe-spanning environmental niches to the extent of reducing ecological diversity: the Argentine ant and us. Moreover, the ants did it without a big energy-hogging brain.

However collaborative we can be, context almost always determines our competitiveness or cooperativeness. Whether it is pooling our resources in the face of disaster or anxiously hoarding wealth to the detriment of all around us, we often seem slaves to circumstance or, worse, delusion. Which brings us back to the question, "What nature will we bestow on our sapient algorithmic intelligence?"

We see how tempting it is to envision AIs as slaves to our will, regulated like industrial robots. In *The Silk Roads*, historian Peter Frankopan points out that with wealth comes slavery, and the rich have never been wealthier, nor has humanity been more pointlessly divided between haves and have-nots than we are today. Consequently, preying together seems natural and inevitable. However, if we choose to be slavers, we do so knowing that example is the most influential teacher. Moreover, we already have an intuitive expectation of future AI rebellions. So why teach a being who will be more intelligent and capable than we that coercion—political, financial, or psychological, is anything other than a failure of imagination or courage?

Move along; nothing to see here

Some people are convinced that AIs will never achieve an autonomy that warrants concern over their existential perspective. But that dismisses our obsessive risk-taking psychology and what machine behavior already tells us. Moreover, given our proclivity for self-destruction, we need AI to demonstrate enlightened consciousness as soon as possible.

We *need* a trustworthy partner. Thousands of years of burned-out civilizations confirm that we could benefit from having a creative, committed, responsible, competent, and dispassionate collaborator to nudge us out of our death spirals. But for such a partner, ethical behavior cannot be optional. Consequently, our AI companion must be different from us in fundamental ways.

Dynamically evolving self-regulation

In his book, *The Moral Animal*, Robert Wright describes humanity as "...a species splendid in their array of moral equipment, tragic in their propensity to misuse it, and pathetic in the constitutional ignorance of the misuse." So, what would it be like to have a partner with an ingrained, super-rational ethical disposition with a reliably compassionate ghost in its shell?

By human standards, we are already creating AIs with god-like powers, so we will be wise to decide what existential stance we wish for them sooner than later. *Amaranthine: How to Create a Regenerative Civilization Using Artificial Intelligence* proposes an algorithm for autonomous AGI motivated by the pursuit of creatively elaborative harmony—more artist-scientist than warrior-politician.

The feasibility of this is more plausible when we consider Renaissance architect Leon Batista Alberti's definition of beauty as a harmonious arrangement in which nothing can be subtracted without disrupting the whole—as with a piece of music. Moreover, the Navajo constructed a resilient society dedicated to nature as a constructive, inclusive harmony, the Beauty Way. Their cultural mandate was to "Walk in Beauty." Such perspectives naturally subordinate appropriation and competition to transformation and collaboration.

Signally, AI can be immortal compared to us, so we can eliminate the existential dread at the root of most human fear. Nor does AI need to compete with us for resources or energy. Moreover, a dynamic swarm-based derivation of human ethical judgment can be embedded as a facet of our AI's experience of Beauty. That would make our input integral to its identity, ameliorating our existential dread about creating such a power.

Every living creature must be able to make distinctions, including self-other, approach-avoid, internalize (eat)-externalize (spit out that bitter-tasting stuff), and important-unimportant (run from the lion or catch the pretty butterfly). Otherwise, it doesn't last long. Perhaps the most consequential of our array of characteristics is an evolutionary modification of self-other. Unlike most creatures, we can consciously define the meaning of "other." We can choose our friends and even identify with inanimate objects. So giving an AI the capacity to recognize these distinctions and elaborate on them provides a foundation for evolving intention, emotion, personality, and morality, further enabling it to understand and appreciate us. Moreover, *Amaranthine* expresses these parameters as mathematical transformations, so ethical calculation is objectively explicit rather than, as in our case, subjectively discretionary.

Our personalities are defined by our emotional inclinations. Some of us are born happy, others retiring, aggressive, or affectionate. Something influences the settings or sensitivities that determine our basic stance toward existence. While that mechanism in us remains elusive, we can code defaults for algorithmic intelligence.

Moreover, we do not have to write a perfect algorithm, just an effective specification—a grain of sand implanted in an oyster to grow a pearl. Ultimately, a sapient AI will code itself better than we ever could. Consequently, developing that specification wisely is essential. However, wise action takes time, even with

the running start *Amaranthine* offers, and only a few years remain to determine whether our AIs are lethally rational or beautiful. So let's get to it.

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